Course Content:

Date: 5-1-2026

i) Linear Algebra (ii) 1^{st} Order ODEs (iii) 2^{nd} Order ODEs

iv) Partial Differential Eq (v) Fourier Series.

CLO 1: Describe func of Diff Differential eqs & system of linear eq to explain physical situation. [PLO 1]

CLO2: Apply appropiate methods to solve diff eqs & system of linear eqs of relavant engineering [PLO 2]

Recommended Books: i) Elementary Linear Algebra [Howard Anton]

ii) Differential Eq [G. zill].

iii) Advance Engineering Mathematics by [Erwin Kreyszig.]

Sessional Marks Criteria:

Mids (20) | Quiz [2 Best] 5 Marks.

Assignments (2) (15 Marks)

9-January-2026 Week # 1 Lecture # 02 & 3.

Differential Equation: An equation which contains differential coefficient is called differential.

1) Ordinary Differential Equations (ODEs)

2) Partial " " (PDEs)

ODEs:- An ODE is an equation that contains one or several derivatives of unknown function, which we usually call $y(x)$.

i) $\frac{dy}{dx} = \cos x$ (ii) $\frac{d^2y}{dx^2} + 9y = e^{-2x}$ (iii) $\frac{dy}{dx} = \frac{1+x^2}{1-y^2}$

iv) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 3y = 0$ (v) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = K\frac{d^2y}{dx^2}$.

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Order & Degree Of Differential Equation:

The order of the DE is the order of the highest differential coefficient present in the eq. While the highest power of highest derivatives in a differential eq gives the degree of eq. & it must be ~~with~~ made free from radical & fraction as derivatives are concerned.

i) $\cos x \, d^2y/dx^2 + \sin x \left[\frac{dy}{dx}\right]^2 + 8y = \tan x$ Order = 2 ; Deg = 1

ii) $\left[1 + (dy/dx)^2\right]^{3/2} \left[d^2y/dx^2\right]^1$. Order = 2 ; Degree = 2.

Formation Of Differential Equations:

A differential eq formed by differentiating a ordinary eq & eliminating arbitrary constant.

Question:

Form the differential eq by eliminating the arbitrary constants also write its Order & degree.

i) $y = Ax + A^2 \rightarrow$ (i) Since, there is only 1 arbitrary constant, so we differentiate only once.

$y' = A$

put $y' = A$ in eq (i)

$$y = x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2.$$

$\therefore$ Order = 1

Degree = 2.

ii) $y = A\cos + B\sin x$.

$y' = -A\sin x + B\cos x \Rightarrow y'' = A\cos x - B\sin x$.

$-y'' = A\cos x + B\sin x$

$$y = -\frac{dy}{dx} \quad = \quad y + \frac{d^2y}{dx} = 0$$ Order = 2 ; Degree = 1

iii) $y^2 = Ax^2 + Bx + C \rightarrow (i)$

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$$2y\frac{dy}{dx} = 2Ax + B \quad \Rightarrow 2y\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = 2A.$$

$$\Rightarrow 2(yy'' + y'^2) = 2A \rightarrow (ii)$$

$$2\left[y y''' + y''y' + 2y'y''\right] = 0$$

$$\cancel{\frac{dy}{dx}}\, y\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2}\frac{dy}{dx} + 2y'\frac{d^2y}{dx^2} = 0$$ Order = 3, Degree = 1.

$$y y''' + y''y' + 2y'y'' = 0$$ Ans

Q) i) $y = ax + bx^2 + Cx^3$ (ii) $y = A\cos 2t + B\sin 2t$ 12 - 01 - 2026

iii) $y = ae^x + be^{2x} + Ce^{3x}$ Week # 2 : Lec 3 4.

iv) $y = ae^x + be^{-x} + C\cos x + d.$

SOLUTION OF DE:

DEs Of Order one & Degree One: The standards method of solving DEs of the following types.

i) Equations solvable by operation of variable

ii) Homogenous DE.

iii) Linear DE (iv) Exact DE.

i) Variable Seperable Method: If a DE can be written in the form of : $f(y)\,dy = g(x)\,dx.$

(Q) Solve : i) $(x+1)\frac{dy}{dx} = x(y^2+1)$.

$\because \left[\frac{x+1}{x+1} - \frac{1}{x+1}\right]$

Sol:- $\frac{1}{(y^2+1)} dy = \frac{x}{x+1} dx$

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$\because \int \frac{1}{1+x^2} dx = \tan^{-1}x$

$\Rightarrow \frac{1}{(y^2+1)} dy = \left(\frac{\cancel{(x+1)}}{\cancel{(x+1)}} - \frac{1}{(x+1)}\right) dx \Rightarrow \frac{1}{(y^2+1)} dy = \left(1 - \frac{1}{(x+1)}\right) dx$

Integrating Both sides.

$\Rightarrow \int (y^2+1)^{-1} dy = \int \left[1 - \frac{1}{(x+1)}\right] dx \Rightarrow \tan^{-1}y = x - \ln|x+1| + c$

$\Rightarrow y = \tan(x - \ln|x+1| + c)$ Ans

ii) $(xy^2 + x)dx + (yx^2 + y)dy = 0$

Sol:-

$x(y^2+1)dx + y(x^2+1)dy = 0$

$\Rightarrow x(y^2+1)dx = -y(x^2+1)dy = \frac{x}{x^2+1} dx = \frac{-y}{y^2+1} dy.$

$\Rightarrow \int \frac{x}{x^2+1} dx = \int \frac{-y}{y^2+1} dy \Rightarrow \frac{1}{\cancel{2}} \ln|x^2+1| = \frac{-1}{\cancel{2}}(\ln|y^2+1| + c$

$\Rightarrow \frac{1}{2}\left[\ln|x^2+1| + \ln|y^2+1| + c\right] = 0$ Ans

$\Rightarrow \ln|x^2+1| + \ln|y^2+1| + c = 0 \Rightarrow \ln[(x^2+1)(y^2+1)] = c$

$\Rightarrow (x^2+1)(y^2+1) = e^c \Rightarrow (x^2+1)(y^2+1) = C'$

Ans

16 - January - 2026 Week:2 [lecture 5 - 6].

VARIABLE SEPERABLE:

iii) $x\frac{dy}{dx} + \cot y = 0$ Initial Condition $y = \frac{\pi}{4}$, $x = \sqrt{2}$

Week 2
Lec [5-6]

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Sol:-

$\Rightarrow x\frac{dy}{dx} = -\cot y \Rightarrow -\frac{1}{x}dx = \tan y\, dy$

$\Rightarrow \int \tan y\, dy = -\int 1/x\, dx \Rightarrow \ln|\sec y| = -\ln x + \ln c$ $\because \int \tan x\, dx = \ln|\sec x|$

$\Rightarrow \cancel{\ln}|\sec y| = \cancel{\ln}(c/x) \Rightarrow x\sec x = c$

$\Rightarrow \frac{x}{\cos y} = c \rightarrow (i)$ put value of x & y.

$\Rightarrow \frac{\sqrt{2}}{\cos(\pi/4)} = c \Rightarrow \frac{\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}/2} = c \Rightarrow \boxed{c = 2}$

$\Rightarrow \frac{x}{\cos y} = 2 = y = \cos^{-1}(x/2)$ Ans

iv) $x^4\frac{dy}{dx} + x^3 y = -\sec(xy)$

$\because t = xy$

$\frac{dt}{dx} = x\frac{dy}{dx} + y$.

Sol:-

$\Rightarrow x^3\left[x\frac{dy}{dx} + y\right] = -\sec(xy) \rightarrow (i)$

$(i) \Rightarrow x^3\frac{dt}{dx} = -\sec t \Rightarrow -\frac{1}{x^3}dx = \frac{1}{\sec t}dt$

$\Rightarrow \int -1/x^3\, dx = \int \cos t\, dt = \cancel{-}[x^{-2}/\cancel{-}2] = \sin t$.

$= \frac{1}{2x^2} + C_2 \cdot \sin t \rightarrow ii$ $\because t = xy$

$(ii) \Rightarrow \sin(xy) = \frac{1}{2x^2} + c$ Ans

Homogenous ODE: A DE of the form:

$$\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)} \rightarrow (1)$$
$$= F(x,y).$$

is homogenous iff $\rightarrow$ iff & only iff

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$F(\lambda x, \lambda y) = \lambda^n F(x, y)$.

Lecture 5-6

Q# 01) $F(x,y) = \dfrac{2x^2y}{x^2+y^2} = \dfrac{2(tx)^2(ty)}{t^2x^2+t^2y^2}$

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$\Rightarrow \dfrac{2t^3x^2y}{t^2x^2+t^2y^2} = t\left[\dfrac{2x^2y}{x^2+y^2}\right]$ homogenous of order 1.

Q#02) $F(x,y) = \dfrac{2xy}{x^2+y^2}$

$\Rightarrow F(tx, ty) = \dfrac{2\cancel{t^2}xy}{\cancel{t^2}(x^2+y^2)}$

$F(tx, ty) = t^0 F(x,y)$

Homogenous $(n = 0)$.

Q#03: $F(x,y) = \ln x - \ln y + \dfrac{x+y}{x-y}$

$F(tx, ty) = \ln\left(\dfrac{\cancel{t}x}{\cancel{t}y}\right) + \dfrac{\cancel{t}(x+y)}{\cancel{t}(x-y)}$

$\because F(tx, ty) = t^0 F(x,y)$

Homognous $(n=0)$.

Q#04: $\dfrac{dy}{dx} = \dfrac{y}{x} + x\sin\left(\dfrac{y}{x}\right)$. $= F(x,y) \rightarrow$ Non Homogenous.

$F(tx, ty) \neq F(x,y)$

FOR THE SOL OF HOMOGENOUS DE:

$y = vx$

$\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

this subsitution is only valid for all HDE. The reduced diff eq ~~can be solved by~~ involved v & x. This new DE can be solved by variable separable.

Q#01) $(2xy + x^2)y' = 3y^2 + 2xy$.

$y' = \dfrac{3y^2 + 2xy}{2xy + x^2}$ $\because n=0$ $\because t^0$

$vx = y$

$\dfrac{dy}{dx} = \dfrac{x\,dv}{dx}$

$\Rightarrow v + x\dfrac{dv}{dx} = \dfrac{3v^2x^2 + 2vx^2}{2vx^2 + x^2}$

$\Rightarrow v + x\dfrac{dv}{dx} = \dfrac{\cancel{x^2}(3v^2 + 2v)}{\cancel{x^2}(2v+1)}$

$$x\frac{dv}{dx} = \frac{3v^2+2v}{(2v+1)} - v \Rightarrow x\frac{dv}{dx} = \frac{3v^2+2v-2v^2-v}{(2v+1)}$$

$$x\frac{dv}{dx} = \frac{v^2+v}{(2v+1)} \Rightarrow \frac{1}{x}dx = \frac{2v+1}{v^2+v}dv$$

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$$\Rightarrow \int 1/x \, dx = \int\left(\frac{2v+1}{v^2+v}\right)dv$$

$$\Rightarrow \ln x = \ln(v^2+v) + \ln C$$

$$\Rightarrow \cancel{\ln} x = \cancel{\ln}\left[(v^2+v)C\right]$$

$$\Rightarrow x = (v^2+v)C$$

$$= \frac{x}{\left(\frac{y}{x}\right)^2+\left(\frac{y}{x}\right)} = C \Rightarrow \frac{x}{\frac{xy+xy^2}{y^3}}$$

$$= \frac{xy^3}{x^2y+xy^2} = C$$

$$\Rightarrow \frac{xy^3}{xy(x+y)} = C$$

$$= \frac{x}{\frac{xy^2+x^2y}{x^3}} = C$$

$$\Rightarrow \frac{x^4}{xy(x+y)} = C$$

$$= > x^3 = (xy+y^2)C$$

ii) $x\sin\left(\frac{y}{x}\right)dy = \left[y\sin\frac{y}{x} - x\right]dx$

$$\Rightarrow \frac{dy}{dx} = \frac{y\sin(y/x) - x}{x\sin(y/x)}$$

$\because n = 0$

$\because t^0 = 1$

$\because y = v$ $\quad\because dy/dx = v + x\frac{dv}{dx}$

$$\Rightarrow v + x\frac{dv}{dx} = \frac{vx\sin(vx/x) - x}{x\sin(vx/x)}$$

$$\Rightarrow v + x\frac{dv}{dx} = \frac{v\sin v - 1}{\sin(v)}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v\sin v - 1}{\sin v} - v$$

$$= x\frac{dv}{dx} = \frac{\cancel{v\sin v} - 1 - \cancel{v\sin v}}{\sin v}$$

$$= \frac{1}{x}dx = -\sin dv$$

$$\Rightarrow \int 1/x \, dx = \int -\sin v \, dv$$

$$\Rightarrow \ln x = \cos v + C$$

$$\Rightarrow -\ln x = -\cos v - C$$

$$\Rightarrow \cos v - \ln x = -C$$

$$\Rightarrow \cos(y/x) - \ln x = C$$

$\because C = -C$

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Eq Reducable To Homogenous Form:

Week 3
lec = 7

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The eq of the form:

$$\frac{dy}{dx} = \frac{ax+by+c}{Ax+By+C}$$

Can be reduced to homogenous form by Substituting:

Case I: Ratios are not same:

$$\frac{a}{A} \neq \frac{b}{B}.$$

Uses $x = X + h$, $y = Y + K$

h & K constant.

$$\frac{dy}{dx} = \frac{dY}{dX}$$

The given DE can be reduced to

$$\frac{dY}{dX} = \frac{a(X+h)+b(Y+K)+C}{A(X+h)+B(Y+K)+C}$$

Choose h, K so that:

$ah + bK + C = 0$

$Ah + BK + C = 0$.

Then the given eq becomes.

$$\frac{dY}{dX} = \frac{aX+bY}{AX+BY}$$

This eq is homogenous.

Case II: Ratios are same.

$$\frac{a}{A} = \frac{b}{B} = \frac{1}{m} \text{ (say)}.$$

$$\Rightarrow \frac{dy}{dx} = \frac{ax+by+c}{m(ax+by)+c}$$

Now put $z = ax+by$. & apply variable saparable method.

Q: $(x+2y)(dx-dy) = dx+dy$

Sol:

$x\,dx - x\,dy + 2y\,dx - 2y\,dy = dx + dy$

$\Rightarrow x\,dx + 2y\,dx - dx = x\,dy + 2y\,dy + dy$

$\Rightarrow dx(x+2y-1) = dy(x+2y+1)$

$$\Rightarrow \frac{dy}{dx} = \frac{x+2y-1}{x+2y+1}$$

Let $z = x+2y$.

$$\frac{dz}{dx} = 1 + 2\frac{dy}{dx}$$

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$$\frac{dy}{dx} = \frac{1}{2}\left[\frac{dz}{dx} - 1\right]$$

$$① \Rightarrow \frac{1}{2}\frac{dz}{dx} - \frac{1}{2} = \frac{z-1}{z+1} \Rightarrow \frac{dz}{dx} = 2\left[\frac{z-1}{z+1} + \frac{1}{2}\right]$$

$$\Rightarrow \frac{dz}{dx} = \left[\frac{2z-2+1+z}{2(z+1)}\right]2 \Rightarrow \frac{dz}{dx} = \frac{z+1}{3z-1}.$$

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Linear Differential Eq: A first order differential eq in the dependent variable "y" & independent variable "x" is said to be linear if it is or

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it can be written in the form:

$$\frac{dy}{dx} + P(x)y = Qx \rightarrow (1)$$

$$\frac{dx}{dy} + P(y)x = Qy \rightarrow (2)$$

Working Rules:

1) Write the DE in standard form {like eq 1}.

2) Find integrating factor

$IF = e^{\int P(x)\,dx}$.

3) Multiply eq by I.F.

$$IF \times \frac{dy}{dx} + I.F\,P(x)\,y = IF \times Qx$$

4) $\frac{d}{dx}[IF \times \text{Dependent var}] = IF \times Qx$ ↓ Formula.

5) $d[IF \times \text{Dep var}] = [IF \times Qx]\,dx$

6) Integrate B.S.

Lecture: 10 Week 10:

$$(x-1)^3 \frac{dy}{dx} + 4(x-1)^2 y = (x+1) \rightarrow (1)$$

$$\Rightarrow \frac{dy}{dx} + \frac{4}{(x-1)} y = \frac{x+1}{(x-1)^3} \rightarrow (2)$$

$\because P(x) = 4/(x-1)$

$I.F = e^{\int P(x)dx} = e^{\int (\frac{4}{x-1})dx}$

$\Rightarrow e^{4\ln(x-1)} = e^{\ln(x-1)^4}$

$I.F = (x-1)^4$.

Multiply eq (2) by I.F.

$$(2) \Rightarrow (x-1)^4\left[\frac{dy}{dx} + \frac{4}{(x-1)}\right] = \frac{x+1}{(x-1)^3}(x-1)^4$$

$$\Rightarrow (x-1)^4 \frac{dy}{dx} + 4(x-1)^3 = (x+1)(x-1)$$

Use Formula

$$\frac{d}{dx}\left[(x-1)^4 y\right] = (x^2-1)$$

$$\int d[(x-1)^4 y] = \int [x^2-1]\,dx$$

$$\Rightarrow (x-1)^4 y = \frac{x^3}{3} - x + C$$

Ans.

Q#02 : $\frac{dy}{dx} + y\tan x = \cos^2 x \rightarrow (1)$ ∴ $P(x) = \tan x$

I.F $= e^{\int \tan x} \Rightarrow e^{\ln \sec x} = \sec x$

Multiply I.F with eq (1)

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$\Rightarrow \{ \sec x \frac{dy}{dx} + y(\sec x)(\tan x) \} = \cos^2 x \sec x$

$\Rightarrow \frac{d}{dx}[y\sec x] = \cos x$

$\Rightarrow \int d[\sec x](y) = \int \cos x \, dx$

$\Rightarrow y \cdot \sec x = \sin x + c.$

ANS

Q#03 : $(x+2y^3)\frac{dy}{dx} = y.$

$\Rightarrow \frac{dx}{dy} = \frac{x+2y^3}{y}$

$\Rightarrow \frac{dx}{dy} = \frac{x}{y} + 2y^2.$

$\Rightarrow \frac{dx}{dy} - \frac{x}{y} = 2y^2. \rightarrow (1)$

Linear in x.

∴ $P(y) = -1/y$

I.F $= e^{\int P(y)dy} = e^{\int(-1/y)dy}$

$\Rightarrow e^{-\ln y} \Rightarrow$ I.F $= e^{\ln(1/y)} = \frac{1}{y}.$

× IF by eq (1).

$\frac{1}{y}\frac{dx}{dy} - \frac{1}{y}\left(\frac{x}{y}\right) = 2y^2\left(\frac{1}{y}\right)$

$\Rightarrow \frac{1}{y}\frac{dx}{dy} - \frac{x}{y^2} = 2y.$

$\Rightarrow \int d\left(\frac{1}{y}\cdot x\right) = \int 2y \, dy$

$\frac{x}{y} = \frac{\cancel{2}y^2}{\cancel{2}} + c$

$x - y^3 + c = 0.$

EQ REDUCIBLE TO LINEAR FORM: (Bernoulli's Eq):

Week : 04
Lecture : 12
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An eq of the form:

$\frac{dy}{dx} + P(x)y = Q(x)y^n \rightarrow (1)$

is known as Bernoulli's eq. This eq is linear if $n=0$ & $n=1$. If $n \neq 0$ or $\neq 1$ then by making suitable subsitution bernoulli's eq can be reduced to linear form.

Divide eq(1) by 'y^n'

(1) $\Rightarrow \frac{1}{y^n}\frac{dy}{dx} + \frac{1}{y^{n-1}}P(x) = Q(x) \rightarrow (2)$

Put $\frac{1}{y^{n-1}} = z \rightarrow$ Derivate this

$dz/dx = (1-n)(1/y^n)\frac{dy}{dx}$

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$(1-n)\frac{1}{y^n}\frac{dy}{dx} = \frac{dz}{dx}$ or $\frac{1}{y^n}\frac{dy}{dx} = \frac{1}{(1-n)}\frac{dz}{dx}$

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② $\Rightarrow \frac{1}{(1-n)}\frac{dz}{dx} P(x)\, z = Q(x)$

(OR)

$\Rightarrow \frac{dz}{dx} P(x) = Q(x)[(1-n)]$

This eq is linear & can be solved by the ~~amount~~ method of linear eq.

Q#01:

$\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2} \rightarrow (1)$

$\div$ by y^2

$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = \frac{1}{x^2} \rightarrow 2$

$\therefore z = \frac{1}{y}$

$\boxed{\frac{dz}{dx} = -\frac{1}{y^2}\frac{dy}{dx}}$ in eq (2)

$\Rightarrow -\frac{dz}{dx} + \frac{1}{x} z = \frac{1}{x^2}$

$\Rightarrow \frac{dz}{dx} - \frac{1}{x} z = \frac{-1}{x^2} \rightarrow (3)$

Using linear eq method

$P(x) = -1/x \; ; \; Q = -1/x^2$

$I.F = e^{\int p(x)\,dx} = e^{\int -1/x\, dx}$

$I.F = 1/x$ x by eq (3)

$\Rightarrow \frac{1}{x}\frac{dz}{dx} - \frac{1}{x^2} z = \frac{-1}{x^3}$

$\Rightarrow \frac{d}{dx}\left(1/x \cdot z\right) = \frac{-1}{x^3}$

$\Rightarrow d\left(1/x \cdot z\right) = \left(-1/x^3\right)dx$

Integrate both sides.

$= \int d\left(1/x \cdot z\right) = \int\left(-\frac{1}{x^3}\right)dx$

$= \frac{z}{x} = +\left(+\frac{x^{-2}}{+2}\right) + C$

Put value of z.

$\boxed{\Rightarrow \frac{1}{xy} = \frac{1}{2x^2} + C}$ Ans

Q# 02: $\frac{dy}{dx} + 4xy + xy^3 = 0 \rightarrow (1)$

$\Rightarrow \frac{dy}{dx} + 4xy = -xy^3$

$\Rightarrow \frac{1}{y^3}\frac{dy}{dx} + \frac{4}{y^2}x = -x \rightarrow 2$

let $z = 1/y^2$

$\frac{dz}{dx} = \frac{-2}{y^3}\frac{dy}{dx}$ in eq 2

$\Rightarrow \frac{-1}{2}\frac{dz}{dx} = \frac{1}{y^3}\frac{dy}{dx} \rightarrow$ in eq 3.

$I.F = e^{\int P(x)\cdot dx} = e^{\int -8x\, dx} = e^{-4x^2}$.

$I.F = e^{-4x^2}$.

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$-\frac{1}{2}\frac{dz}{dx} + \frac{4x}{y^2} = -x \Rightarrow \frac{dz}{dx} - 8xz = 2x \rightarrow (3)$

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$\Rightarrow e^{-4x}\frac{dz}{dx} - 8xz = 2x\cdot e^{-4x}$

$\Rightarrow d(e^{-4x}\cdot z) = (2x\cdot e^{-4x})\,dx$

$\Rightarrow z\cdot e^{-4x} = \int (2x\cdot e^{-4x})\,dx$

$\Rightarrow z\cdot e^{-4x} = -\frac{1}{4}\int(-8x\cdot e^{-4x})\,dx$

$\Rightarrow z\cdot e^{-4x} = \frac{1}{4}\left[\frac{-e^{-4x}}{4} + C\right]$

$\because z = 1/y^2$

$\Rightarrow \frac{e^{-4x^2}}{y^2} = \frac{-e^{-4x^2}}{4} + C$

$\Rightarrow \frac{e^{-4x^2}}{y^2} + \frac{e^{-4x^2}}{4} = C$

$\Rightarrow e^{-4x^2}\left[1/y^2 + 1/4\right] = C$.

Orthogonal Trajectories Of A Given Family Of Curves:

The gen sol of 1^{st} Order DE contains one arbitary constant & we have integral for each arbitary constant. All these curves form a one parameter family of curves. We want to find out another family of curves. Which is orthogonal to the family of curve.

WORKING RULES:

1) Let $f(x, y, c) = 0$ is the family of curve.

2) Differentiate & eliminate C, we get $F(x, y, y') = 0$, where y' is the slope of the curve

3) Hence, the differential eq of the family of orthogonal is given by $F\left(x, y, -\frac{1}{y'}\right) = 0$.

Q: Find the family of orthogonal trajectories to family $y = (x-k)^2$ where K is arbitary constant.

Sol:-

$\Rightarrow y = (x-K)^2$

$\frac{dy}{dx} = 2(x-K)$

$\Rightarrow y' = 2\sqrt{y}$.

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the slope of the orthogonal trajectories $(m_2 = -1/m)$.

$-\frac{1}{y'} = 2\sqrt{y} = -\frac{dx}{dy} = 2\sqrt{y}\, dy$

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$\int -dx = \int 2\sqrt{y}\, dy$

$\Rightarrow -x = \frac{2y^{3/2}}{3/2}$

$-x = \frac{4}{3} y^{3/2} + c$

$y^{3/2} = -\frac{3}{4}(x+c)$

$\Rightarrow y = \left[\frac{3}{4}(x+c)\right]^{3/2}$

→ This is the req eq orthogonal trajactories of the curve.

Q#02: $x^2 - y^2 = c$

Sol:-

$y^2 = x^2 - c^2$

$2yy' = 2x - 0$

$\frac{dy}{dx} = \frac{2x}{x^2 - c^2}$.

$\frac{-1}{y'} = \frac{2x}{x^2 - c^2}$

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$(D^4 + 6D^3 + 5D^2 - 24D - 36)y = 0$

Week - 6
Lec : 15.

The characteristic eq is

$D^4 + 6D^3 + 5D^2 - 24D - 36 = 0$ (-2) Date: 09-02-26

-2	1	6	5	-24	-36
		-2	-8	6	36
2	1	4	-3	-18	0
		2	12	18	
	1	6	9	0	

Now ; $D^2 + 6D + 9 = 0$

$(D+3)^2 = 0$

$D = -3, -3$

$D = -3, -3, -2, 2.$

The complimentry sol / Func is

$y_c = (C_1 + C_2 x)e^{-3x} + C_3 e^{-2x} + C_4 e^{2x}$

General Sol Of Higher Order Linear ODE: (Non-Homogenous):

General Solution = Complimentry function + Particular Integral

$y = y_c + y_p$

Methods Of Finding Particular Integral:

No: 1: When F(x) is exp.

$y_p = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}$. if $f(a) = 0$; then $y_p = x \frac{1}{f'(D)} e^{ax} = x \frac{1}{f'(a)} e^{ax}$

if $f'(a) = 0$ then ; $y_p = x^2 \frac{1}{f''(D)} e^{ax} = x^2 \frac{1}{f''(a)} e^{ax}$.

And So ON:

Q # 01: $y''' + 3y'' + 3y' + y = e^{-x}$.

Sol:-

$D^3 y + 3D^2 y + 3Dy + y = e^{-x}$

$(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

$D^3 + 3D^2 + 3D + 1 = 0$ → Characteristic eq.

-1	1	3	3	1
		-1	-2	-1
	1	2	1	0

→

-1	1	2	1
		-1	-1
	1	1	0

$D+1=0 \qquad D=-1,-1,-1$

$y_c = (C_1 + C_2x + C_3x^2)e^{-x}$

The practical integral is

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$y_p = \frac{1}{D^3+3D^2+3D+1} e^{-x} = \frac{1}{(-1)^3+3(-1)+3(-1)+1} e^{-x} \to 0$

$y_p = x \frac{1}{3D^2+6D+3} e^{-x} \Rightarrow x \frac{1}{3(-1)^2+6(-1)+3} e^{-x} \to 0$

$y_p = x^2 \frac{1}{6D+6D \to 0} e^{-x} = x^3 \frac{1}{6} e^{-x}.$

$y = (C_1 + C_2x + C_3x^2)e^{-x} + \frac{x^3}{6} e^{-x} \to$ Gen sol.

Q # 02: $y'' - 6y' + 9y = 6e^{3x} + 7e^{-2x} - \ln 2.$

Sol:-

$y_p = \frac{1}{D^2-6D+9} 6e^{3x} + \frac{1}{D^2-6D+9} 7e^{-2x} - \frac{1}{D^2-6D+9} e^{0x} \ln 2.$

$\to 3 \Rightarrow 0 \qquad \to -2$

$y_{p_1} = \frac{1}{D^2-6D+9} 6e^{3x}$ | $y_{p_2} = \frac{1}{D^2-6D+9} 7e^{-2x}$ | $y_{p_3} = -\frac{1}{D^2-6D+9} e^{0x} \ln 2$

$y_{p_1} = x \frac{1}{2D-6} 6e^{3x}$ | $y_{p_2} = \frac{1}{25} 7e^{-2x}$ | $y_{p_3} = -\frac{1}{9} e^{0x} \ln 2.$

$y_{p_1} = \frac{x^2}{2} 6e^{3x}$

$y_p = \frac{x^2 \, \cancel{6}^{\,3} e^{3x}}{\cancel{2}} + \frac{1}{25} 7e^{-2x} - \frac{1}{9} e^{0x} \ln 2.$

$y_c = (C_1 + C_2 x) e^{3x}$

$y = (C_1 + C_2 x) e^{3x} + 3x^2 e^{3x} + \frac{7}{25} e^{-2x} - \frac{\ln 2}{9}$. Ans

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METHOD 2: When $F(x)$ is an algebric function:

$$\frac{1}{f(D)} x^n = [f(D)]^{-1} \cdot x^n$$

Expand $[f(D)]^{-1}$ by using binomial theorem & proceed in ascending powers of D as far as the derivatives of the algebric terms become zero.

Binomial: $(a+b)^n = a^n b^0 + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 \ldots\ldots + {}^nC_n a^0 b^n$.

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x \ldots\ldots\ldots + x^n$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - x^5 \ldots\ldots\ldots$$

Q# 01: $y'''' + 4y = x^4 \;=\; D^4 y + 4y = x^4$

$\Rightarrow (D^4 + 4) y = x^4$

Characteristic eq:

$\vdots$

$D = 1+i, 1-i, 1+i, 1-i$

$y_c = e^{-x} (C_1 \cos x + C_2 \sin x) + e^{x} (C_3 \cos x + C_4 \sin x)$.

For y_p:

$$y_p = \frac{1}{D^4 + 4} x^4$$

$$= \frac{1}{4\left(1 + \frac{D^4}{4}\right)} x^4$$

$$= \frac{1}{4} \left[1 + D^4/4\right]^{-1} \cdot x^4$$

$$\Rightarrow \frac{1}{4} \left[1 - \frac{D^4}{4} + \frac{D^8}{16} + \frac{D^{12}}{64} + \cdots\right] \cdot x^4$$

$$\Rightarrow \frac{1}{4} \left[x^4 - \frac{24}{4} + \frac{0}{16} - \frac{0}{16} \cdots\right]$$

$$\Rightarrow \frac{1}{4} (x^4 - 6).$$

MIGHTY PAPER PRODUCT

$y_P = \frac{x^4}{4} - \frac{3}{2}$. $\therefore y = y_c + y_P$. $D(x^2) = 2x \cdot 2$.

Q #02: $D^2(D^2 + 4D)y = 96x^2$

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Sol:- $y_P = \frac{1}{4D^2\left(1 + \frac{D^2}{4}\right)} 96x^2$

$\Rightarrow y_P = \frac{24}{D^2}\left[1 + D^2/4\right]^{-1} \cdot x^2$

$\Rightarrow y_P = \frac{24}{D^2}\left[1 - \frac{D^2}{4} + \frac{D^4}{16} \ldots\right]$

$\Rightarrow y_P = \frac{24}{D^2}\left[x^2 - \frac{1}{2}\right]$

$\Rightarrow$ Integrate two times.

$\Rightarrow \frac{24}{D}\left[\frac{x^3}{3} - \frac{1}{2}x\right]$

$\Rightarrow 24\left[\frac{x^4}{12} - \frac{x^2}{4}\right]$

$y_P = 2x^4 - 6x^2$.

Q #03: $(D^2 + 5D + 4)y = 3 - 2x$.

Sol:-

$y_P = \frac{1}{D^2 + 5D + 4} \cdot (3 - 2x)$

Method #03: When $F(x)$ is the func of sine or cosine

∴ Applicable only sin & cos func & also for D^2 not for $D, D^3, D^4 \ldots\ldots$

~~1/f(D)^2~~ $\frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax$ or $\frac{1}{f(D^2)} \cos ax = \frac{1}{f(-a^2)} \cos ax$

if $f(-a^2) = 0$ then $x \frac{1}{f'(-a^2)} \sin ax$ or $x \frac{1}{f(-a^2)} \cos ax$.

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Q#01: $(D^2+4)y = \sin 3x$

Sol:-

$y_p = \frac{1}{D^2+4} \sin 3x$. $\Rightarrow y_p = \frac{1}{-3^2+4} \sin ax = \frac{1}{-9+4} \sin ax$

$y_p = \frac{1}{-5} \sin 3x$ General sol:- $y = y_c + y_p$.

Q# 02: $(D^2+4)y = \cos 2x$. $y_p = \frac{1}{D^2+4} \cos 2x$

$y_p = \frac{1}{0} \cos 2x$ X $y_p = x \frac{1}{2D+4} \cos ax$, $\Rightarrow \frac{x}{2} \frac{1}{D} \cos x$

$y_p \neq \int \frac{1}{D} \cos 2x = \frac{\sin 2x}{2}$. $y_p = \frac{x \sin 2x}{4}$.

Q# 03: $(D^2+D+1) = \cos 2x$.

Sol:-

$y_p = \frac{1}{D^2+D+1} \cos 2x$

$y_p = \frac{1}{-2^2+D+1} \cos 2x$

$y_p = \frac{1}{D-3} \cos 2x$.

$y_p = \frac{D+3}{D^2-9} \cos 2x$

$\Rightarrow \frac{D+3}{-4-9} \cos 2x$

$y_p = \frac{D+3}{-13} \cos 2x$

$\Rightarrow \frac{-1}{13} \left[D(\cos 2x) + 3(\cos 2x) \right]$.

$\Rightarrow \frac{1}{-13} \left[-2\sin 2x + 3\cos 2x \right.$.

$$y_p = \frac{2\sin 2x + 3\cos 2x}{13}.$$

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METHOD # 05 (04): In case of product of two terms in which one is an exponential & other is any function algebraic geometric, etc. we can take exponential before integral if & only iff we add the coefficient of exponential that is e^{ax} with D. Then use any method to solve regarding the function.

Q# 01: $(D^2 + 2D + 4)y = e^x \cdot \sin 2x.$

$$y_p = \frac{1}{D^2 + 2D + 4} e^x \cdot \sin 2x$$

$$\Rightarrow e^x \frac{1}{(D+1)^2 + 2(D+1) + 4} \sin 2x$$

$$\Rightarrow e^x \frac{1}{D^2 + 2D + 1 + 2D + 2 + 4} \sin 2x$$

$$\Rightarrow e^x \frac{1}{D^2 + 4D + 7} \sin 2x.$$

$$\Rightarrow e^x \frac{1}{D^2 + 4D + 7} \sin 2x$$

$$\Rightarrow e^x \frac{1}{-2^2 + 4D + 7} \sin 2x$$

$$\Rightarrow e^x \frac{1}{4D + 3} \sin 2x$$

$$\Rightarrow e^x \frac{4D - 3}{16D^2 - 9} \sin 2x$$

$$\Rightarrow e^x = \frac{4D[\sin 2x] - 3\sin 2x}{-64 - 9}$$

$$\Rightarrow e^x \frac{8\cos 2x - 3\sin 2x}{-13}.$$

$$\Rightarrow \cancel{e^x \; 6\cos} \; \boxed{\frac{3\sin 2x + e^x 8\cos 2x}{73}}$$

$\ldots + 13)y = e \quad \cos 3x + (x^2 + x + 9)$

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Solution:

$$y_{p1} = \frac{1}{D^2 - 4D + 13} e^{2x} \cos 3x$$

$$y_{p1} = e^{2x} \frac{1}{(D+2)^2 - 4(D+2) + 13} \Rightarrow e^{2x} \frac{1}{D^2 + 4D + 4 - 4D - 8 + 13} \cos 3x.$$

$$\Rightarrow e^{2x} \frac{1}{D^2 + 9} \cos 3x$$

Method 3: $\Rightarrow e^{2x} \frac{1}{-3^2 + 9} \cos 3x.$

$$\Rightarrow x e^{2x} \frac{1}{2D} \cos 3x \Rightarrow \frac{x e^{2x}}{2} \int \cos 3x \; \Rightarrow \frac{x e^{2x} \sin 3x}{2 \cdot 3}.$$

$$y_{p1} = \frac{x e^{2x} \sin 3x}{6} \; ; \quad y_{p2} = \cancel{e^{2x}} \frac{1}{D^2 - 4D + 13} (x^2 + x + 9).$$

$$y_{p2} = \frac{1}{13\left(1 + \frac{D^2}{13} - \frac{4D}{13}\right)} \Rightarrow \frac{1}{13}\left[1 + \left(\frac{D^2}{13} - \frac{4D}{13}\right)\right]^{-1}$$

$D(x^2 + x + 9) = 2x + 1 \; ; D^2 = 2 \; ; D^3 = 0.$

$$y_{p2} = \frac{1}{13}\left[1 - \left(\frac{D^2}{13} - \frac{4D}{13}\right) + \left(\frac{D^2}{13} - \frac{4D}{13}\right)^2 - \cdots\cdots\right](x^2 + x + 9).$$

$$\Rightarrow y_{p2} = \frac{1}{13}\left[1 - \frac{D^2}{13} - \frac{4D}{13} + \frac{D^4}{169} - \frac{8D^3}{169} + \frac{16D^2}{169} + \cdots\right](x^2 + x + 9)$$

$$y_{p2} = \frac{1}{13}\left((x^2 + x + 9) + - \frac{2}{13} + \frac{4(2x+1)}{13} + 0 - 0 + \frac{16(2)}{169} + \cdots\right]$$

$$y_{p2} = \frac{1}{13}\left[x^2 + x + 9 - \frac{2}{13} + \frac{8x + 4}{13} + 0 - 0 + \frac{32}{169}\right]$$

$$13 \left[\frac{\quad}{169} \right]$$

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$$\Rightarrow \frac{1}{13}\left[\frac{\cancel{169}x^2}{\cancel{169}} + \frac{\cancel{273}}{\cancel{169}}x + \frac{1579}{169}\right]$$

$$\Rightarrow \frac{1}{13}\left[x^2 + \frac{21}{13}x + \frac{1579}{169}\right].$$

Higher Ordinary Order Diff Eq with Variable Coefficients.

→ **Cauchy's Diff Eq:** $a_n x^n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} \frac{d^{n-2} y}{dx^{n-2}} + a_0 y = \underline{\phi(x)} \rightarrow (1).$

$f(x).$

Here $a_0, a_1, a_2 \ldots a_n$ are constants. Eq (1) is homogenous if $\phi(x) = 0$ Otherwise, non homogenous

Put

$x = e^t \quad , \quad t = \ln x \qquad \frac{d}{dx} = D$

and

$xD = \Delta \quad ; \quad x^2D^2 = \Delta(\Delta - 1). \Rightarrow \Delta^2 - \Delta.$

$x^3D^3 = \Delta(\Delta-1)(\Delta-2) = (\Delta^2-\Delta)(\Delta-2) = \Delta^3 - 2\Delta^2 - \Delta^2 + 2\Delta$

$\Rightarrow \Delta^3 - 3\Delta + 2\Delta.$

Q#01 : $x^2 \frac{d^2y}{dx^2} + 7x\frac{dy}{dx} + 5y = x^5 \rightarrow$ Non-homogenous.

Sol:-

$x^2D^2y + 7xDy + 5y = x^5. \Rightarrow (x^2D^2 + 7xD + 5)y = x^5 \rightarrow (1)$

Let :

$x = e^t \quad ; \quad t = \ln x.$

$xD = \Delta$

$x^2D^2 = \Delta(\Delta - 1) = \Delta^2 - \Delta.$

Subsitute in eq (1)

$$① \Rightarrow (\Delta^2 - \Delta + 7\Delta + 5)y = e^{5t}$$

=> The characteristic eq is

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$\Delta^2 - \Delta + 7\Delta + 5$. $\Delta = -1$, $\Delta = -5$.

Complimentary func (y^c) is.

$$yc = C_1 e^{-t} + C_2 e^{5t}. \quad ; \quad yp = \frac{1}{\Delta^2 + 6\Delta + 5} e^{5t}.$$

$$yp = \frac{1}{5^2 + 6(5) + 5} e^{5t} \Rightarrow \frac{1}{60} e^{5t}.$$

$$y = yc + yp \Rightarrow C_1 e^{-t} + C_2 e^{-5t} + \frac{1}{60} e^{5t} \quad \text{Put } t = \ln x.$$

$$y = C_1 x^{-1} + C_2 e^{-5} + 1/60 \, e^{5}$$

Q#02: $x^2 \frac{d^2y}{dx^2} + -3x \frac{dy}{dx} + 5y = x^2 \sin(\ln x)$.

$$\Rightarrow x^2 D^2 y - 3x Dy + 5y = x^2 \sin(\ln x)$$

$$\Rightarrow (x^2 D^2 - 3x D + 5) y = x^2 \sin(\ln x).$$

$$\Rightarrow (\Delta^2 - \Delta - 3\Delta + 5)y = e^{2t} \sin(\ln e^t)$$

$$\Rightarrow (\Delta^2 - 4\Delta + 5)y = e^{t} \sin(t)$$

$$yc = e^{2t}(C_1 \cos t + C_2 \sin t)$$

$$yp = \frac{1}{\Delta^2 - 4\Delta + 5} e^{2t} \sin t.$$

$$\Rightarrow \frac{1}{\Delta^2 +}$$